

# Towards a Denotational Semantics for Gradual Typing with Higher-Order Store

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## 1 Introduction

In this note, we will explore how to extend the denotational model of gradual typing from our POPL 2025 paper [3] to accommodate languages with higher-order store. Our starting point is the paper by Sterling et al. on denotational semantics for general store and polymorphism via guarded type theory [4].

We proceed in two phases:

1. As a warm-up, we focus first on the *term* model only. That is, we only interpret types and terms. The main point of interest here will be solving the equation for the dynamic type.
2. We then extend the techniques to the full *relational* model, which means defining a notion of weak bisimilarity, strong error ordering, perturbations, and (quasi-)representability.

### 1.1 Goals

## 2 The Ambient Guarded Type Theory

Before describing the model, we need to fix an ambient guarded metatheory in which we will carry out our constructions. The model in the POPL 2025 paper is constructed in Clocked Cubical Type Theory (CCTT) as introduced by Kristensen et al. [2]. CCTT is a guarded type theory combining the existing notion of *multi-clock guarded recursion* [1] with higher-inductive types and univalence from cubical type theory.

The crucial property about the ambient type theory is that it is compatible with our definition of weak bisimilarity on the free error domain. More specifically, in order for weak bisimilarity on  $L_{\mathcal{U}}A$  to be a *relation*, it is necessary to use an existential quantifier in defining the cases where one side returns immediately and the other side steps. (If we use  $\Sigma$  instead, then the resulting definition is *not* a relation as it is not Prop-valued.) However, one can show that in the topos of trees model of guarded type theory, the diverging computation terminates, if the number of steps is existentially quantified. That is, one can show that  $\exists n. \Omega = \delta^n(\eta x)$  holds, where  $\Omega = \text{fix } \theta$  and  $\delta = \theta \circ \text{next}$ . As a result, under our definition of weak bisimilarity we have  $\Omega \approx (\eta x)$  in the topos of trees model.

The practical consequence of this fact is that our construction of a guarded denotational semantics for gradual typing cannot be carried out in the topos of trees. Fortunately, this issue does not occur in the model of Clocked Cubical Type Theory. That is, one cannot prove in that model that  $\Omega \approx (\eta x)$ .

Things become potentially problematic when we try to reconcile CCTT with the ambient type theory in the paper by Sterling et al. [4]. The key difference in their guarded type theory is the existence of an impredicative universe of types, which plays a crucial role in modeling higher-order store. The authors construct a model of their type theory by essentially taking presheaves on  $\omega$  internally to a realizability topos. This is sufficient for showing the existence of a model of guarded type theory with an impredicative universe. However, it is unclear whether these techniques extend to the more complicated model of Kristensen et al.

This leads us to our first open question:

**Question 1.** *Is it consistent to modify Clocked Cubical Type Theory [2] with an axiom asserting the existence of an impredicative universe of types?*

### 3 The Cast Calculus

We work with a standard gradually-typed lambda calculus with natural numbers and the ability to allocate new reference cells at runtime.

### 4 The Term Model

#### 4.1 Worlds

As in [4], we define worlds via guarded recursion as the solution to

$$\mathbb{W} = \mathbb{N} \multimap_{fin} \triangleright[\mathbb{W}, \mathbb{U}].$$

#### 4.2 CBPV Decomposition

#### 4.3 The Dynamic Type

### 5 Relational Model

#### 5.1 Phase 1: Predomains and Error Domains

#### 5.2 Phase 2: Perturbations and (Quasi-)representable Relations

### 6 Discussion

### References

- [1] Robert Atkey and Conor McBride. Productive coprogramming with guarded recursion. pages 197–208, 2013.
- [2] Magnus Baunsgaard Kristensen, Rasmus Ejlers Mogelberg, and Andrea Vezzosi. Greatest hits: Higher inductive types in coinductive definitions via induction under clocks. 2022.
- [3] Eric Giovannini, Tingting Ding, and Max S. New. Denotational semantics of gradual typing using synthetic guarded domain theory. *Proc. ACM Program. Lang.*, 9(POPL), January 2025.
- [4] Jonathan Sterling, Daniel Gratzer, and Lars Birkedal. Denotational semantics of general store and polymorphism. *CoRR*, 2022.